
반도체 소자 with TCAD

프로젝트 발표 - 7 조

HKMG Split 방안 가이드

 SPROCESS										 SDEVICE	 SVISUAL
	Lg	NWell	HK_Thick	LDD_Dose	LDD_E	Halo_Dose	Halo_E	Halo_Tilt		Metal_WF	Vd

- Lg 0.05, 0.1 0.25, 0.5
- Pwell $1e+15$
- HK_Thick 0.002
- LDD Dose/Energy $5e+13$, 10keV
- Halo Dose $2e+13$
- Halo Energy 20keV

- **Halo Tilt 10~30**
- **Metal Work function 4.5~5.0**

주요 split 변수

Vtgm	Id	SS	gm	t_IL	t_HK	EOT	loff	lg_off	lg_max

제시 조건 : Lg 0.5 기준 Vth 값에 대해 오차범위 10% 만족 / 각 Lg 에 대한 SS < 80 만족

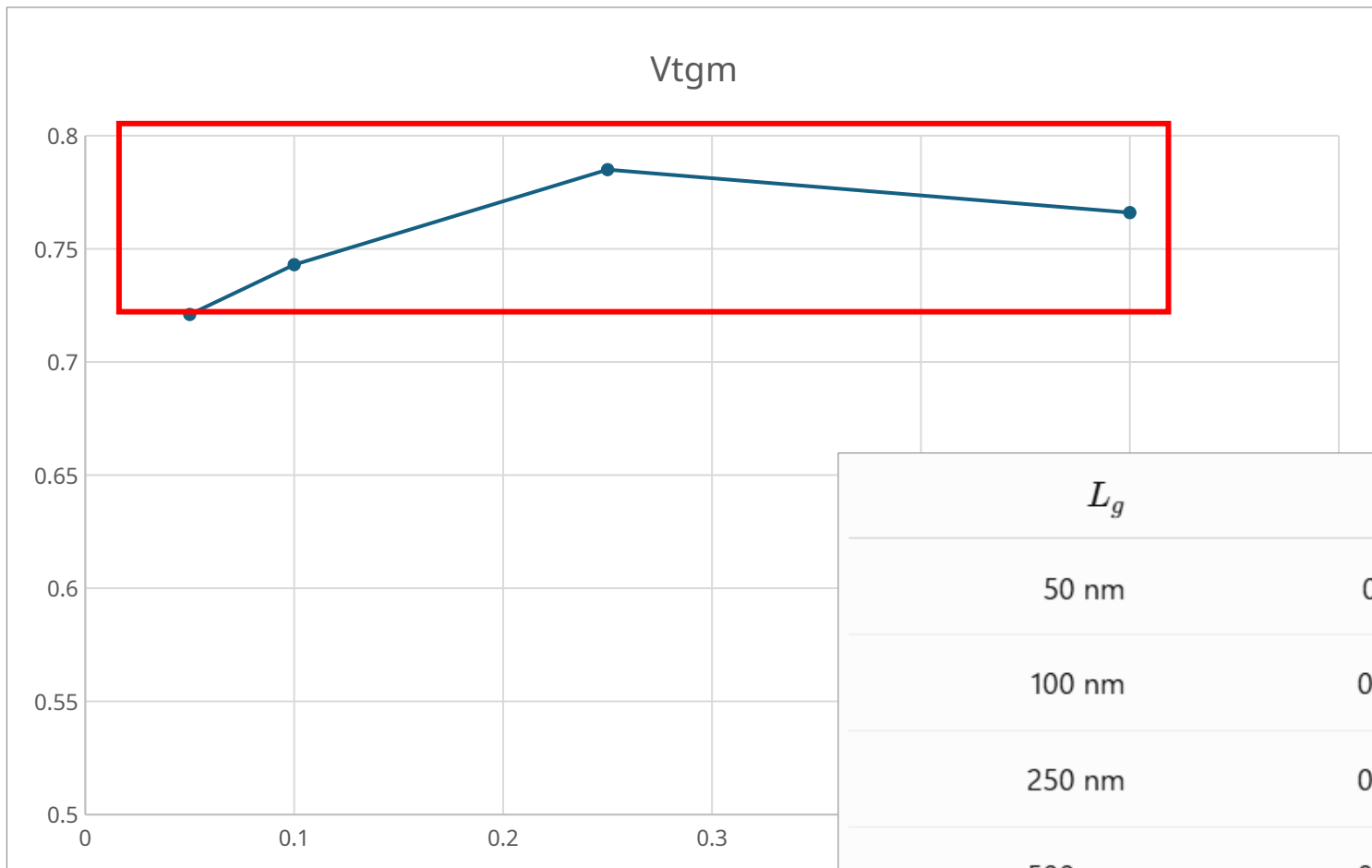
Lg 0.05, 0.1 0.25, 0.5 에 따른 공정 파라미터

주어진 공정 조건 Vth 오차범위 10%
SS < 80 기준을 만족하기 위해
다른 Lg 값과 다르게 설정한 조건

Lg	Pwell	HK_Thick	LDD_Dose	LDD_E	SD_Dose	SD_E	Halo_Dose	Halo_E	Halo_Tilt	Metal_WF
0.05	1e+15	0.002	1e+13	10	9e+14	15	2e+13	20	30	4.5
Lg	Pwell	HK_Thick	LDD_Dose	LDD_E	SD_Dose	SD_E	Halo_Dose	Halo_E	Halo_Tilt	Metal_WF
0.1	1e+15	0.002	5e+13	10	1e+15	15	2e+13	20	30	4.5
Lg	Pwell	HK_Thick	LDD_Dose	LDD_E	SD_Dose	SD_E	Halo_Dose	Halo_E	Halo_Tilt	Metal_WF
0.25	1e+15	0.002	5e+13	10	1e+15	15	2e+13	20	30	4.6
Lg	Pwell	HK_Thick	LDD_Dose	LDD_E	SD_Dose	SD_E	Halo_Dose	Halo_E	Halo_Tilt	Metal_WF
0.5	1e+15	0.002	5e+13	10	1e+15	15	2e+13	20	30	4.6

Device 특성

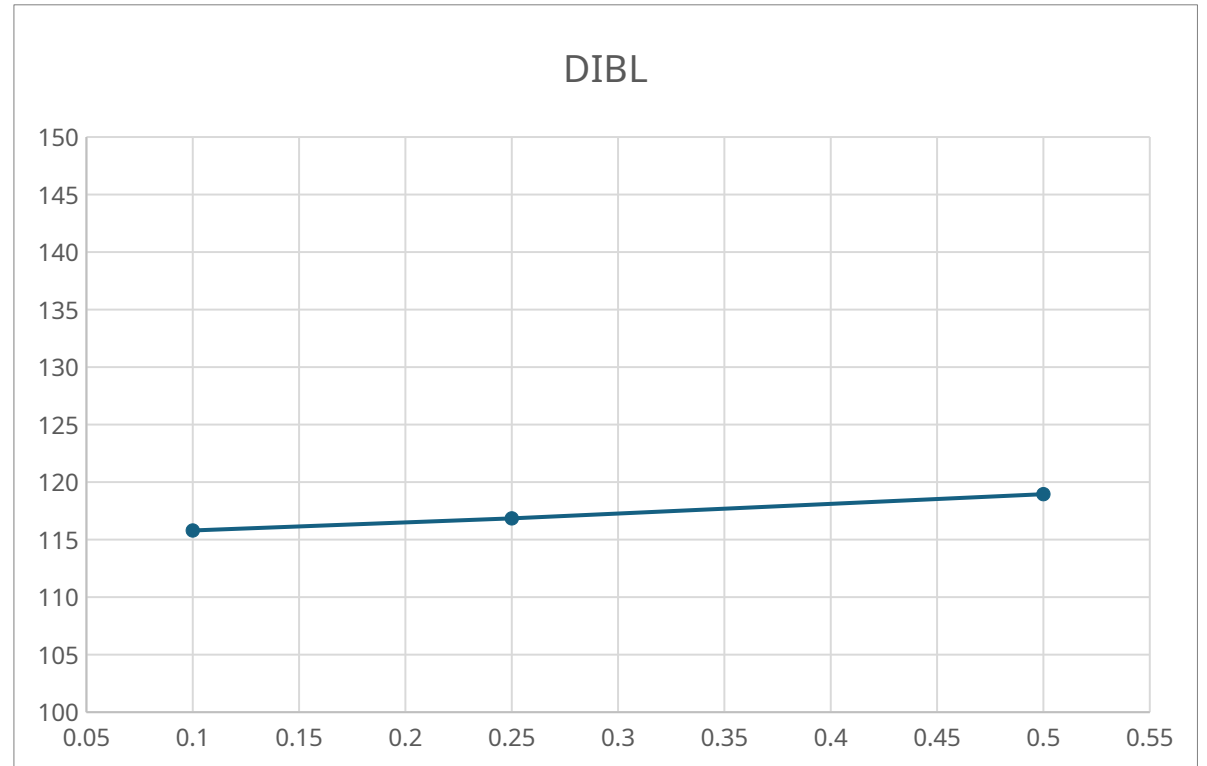
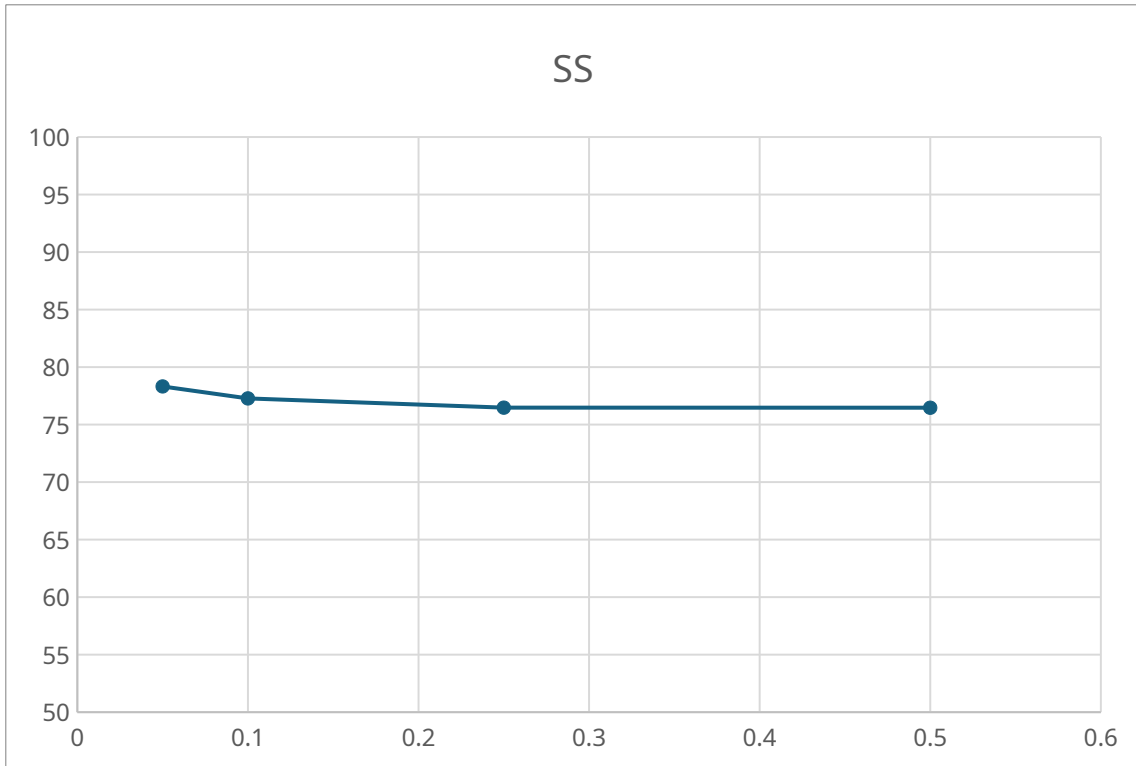
① 소자 특성 Graph



L_g	V_{tgm}	기준 대비 오차	SS
50 nm	0.721 V	5.87%	78.315 mV/dec
100 nm	0.743 V	3.00%	77.270 mV/dec
250 nm	0.785 V	2.48%	76.467 mV/dec
500 nm	0.766 V	기준	76.459 mV/dec

Device 특성

① 소자 특성 Graph

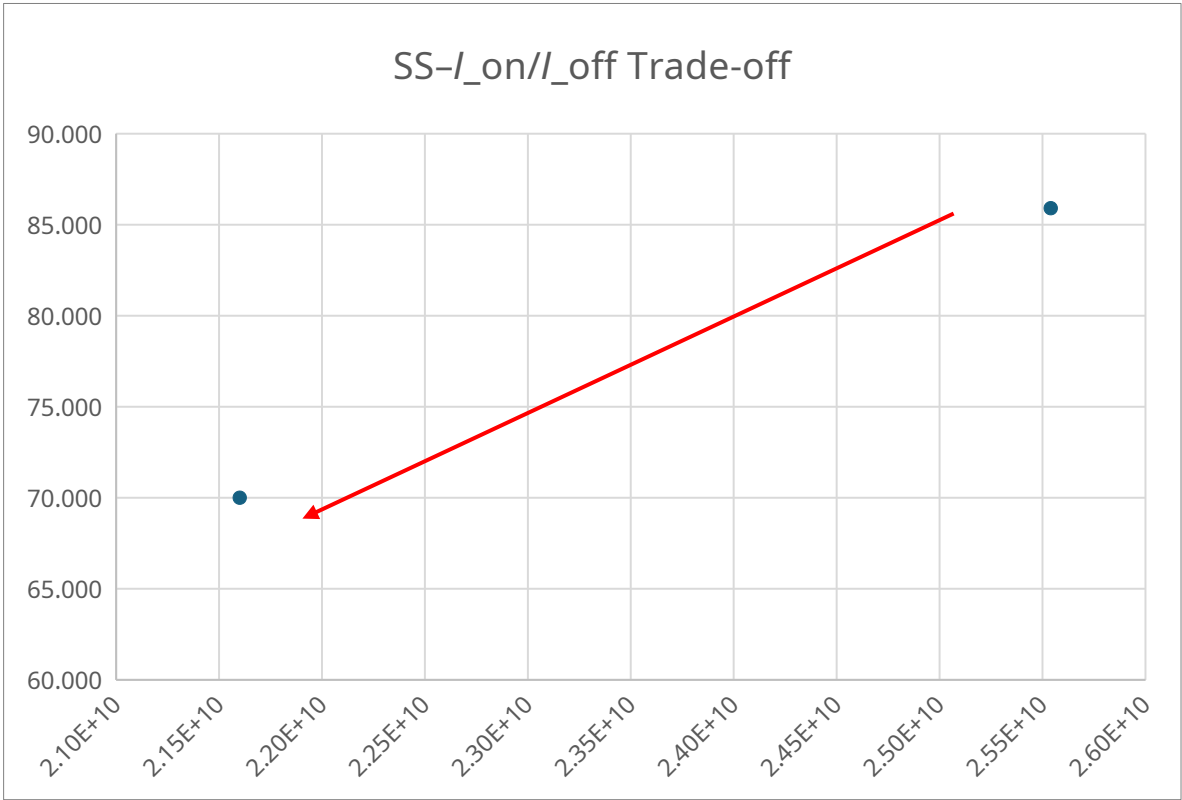


Lg 0.05 제외

최적소자

SimpleMOS vs. Best nMOS

Lg	Halo_Tilt		Metal_WF	Vd		t_IL	t_HK	EOT	Vtgm	Id	Ioff	Ig_off	Ig_max	SS	gm
0.05	30	--	4.5	0.05	--	0.0016	0.0020	0.0019	0.751	1.625e-04	7.523e-15	-5.228e-30	6.875e-22	70.002	3.864e-04
				1.0	--	0.0016	0.0020	0.0019	0.795	4.505e-04	4.785e-14	-9.501e-25	3.725e-22	78.605	1.113e-03

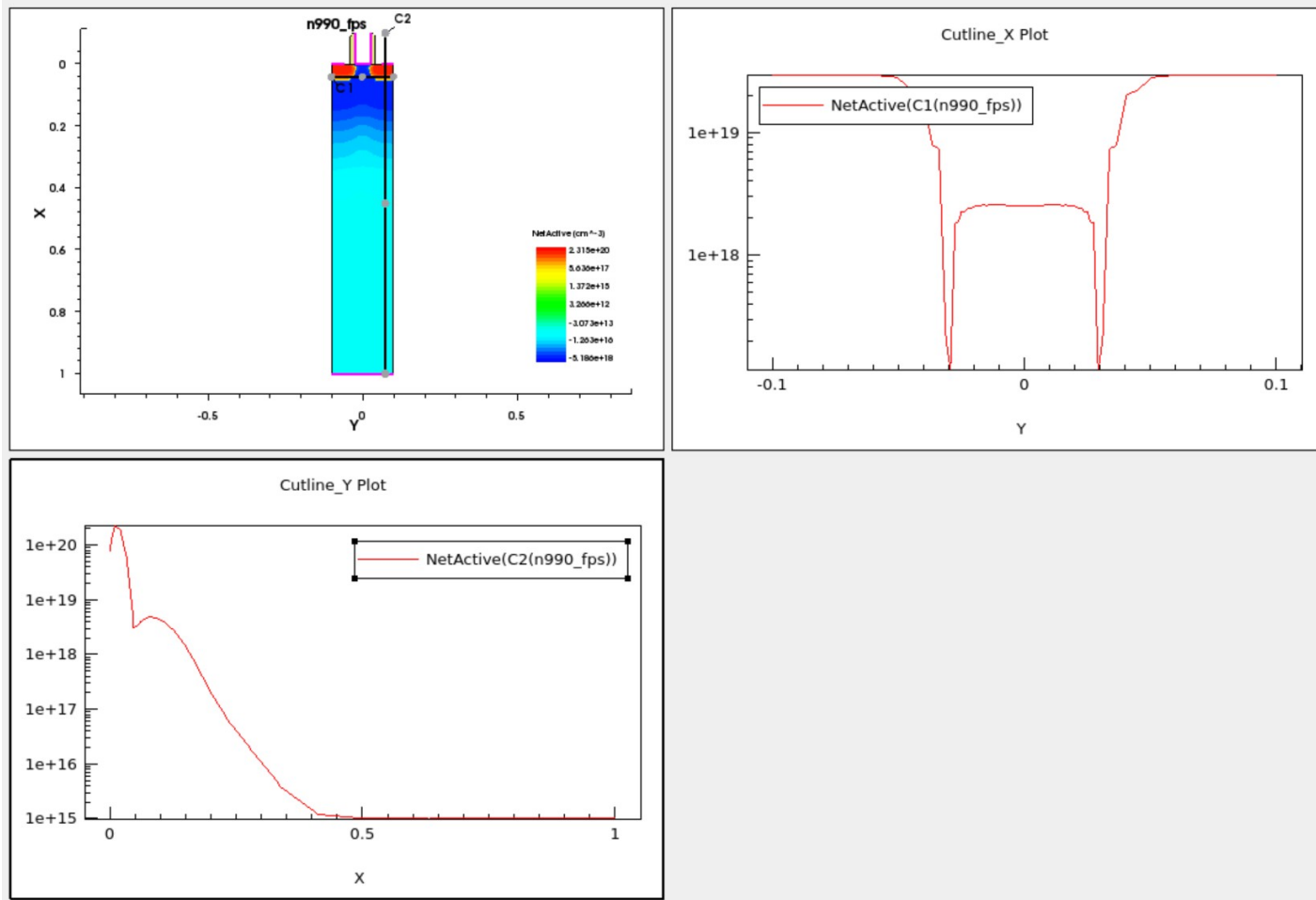


SS가 85.907 mV/dec에서 70.002 mV/dec로 감소
 약 **18.5% 개선**

SS 최적화 과정에서 구동전류 I_{on} 은 증가했지만, 누설전류 I_{off} 가 더 큰 비율로 증가
 → I_{on}/I_{off} 비는 약 15.3% 감소

Device 특성

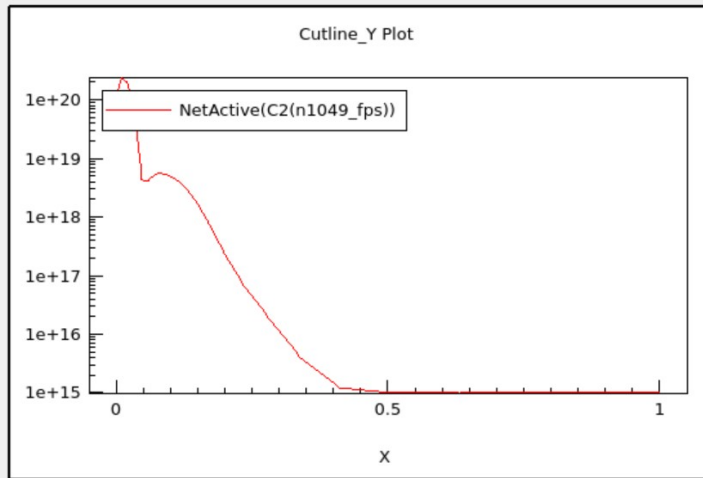
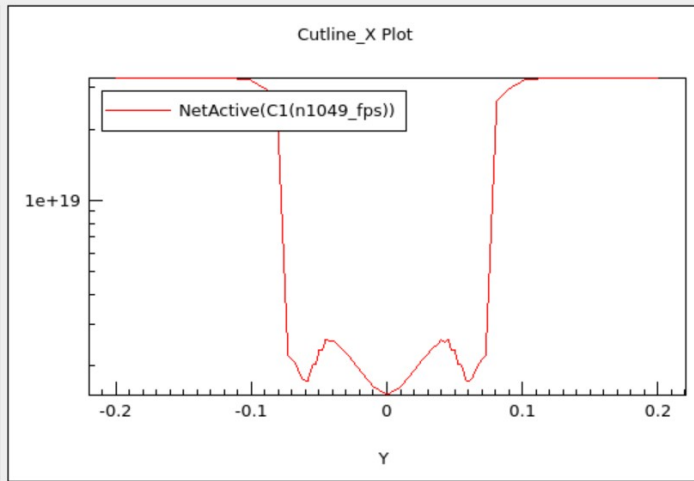
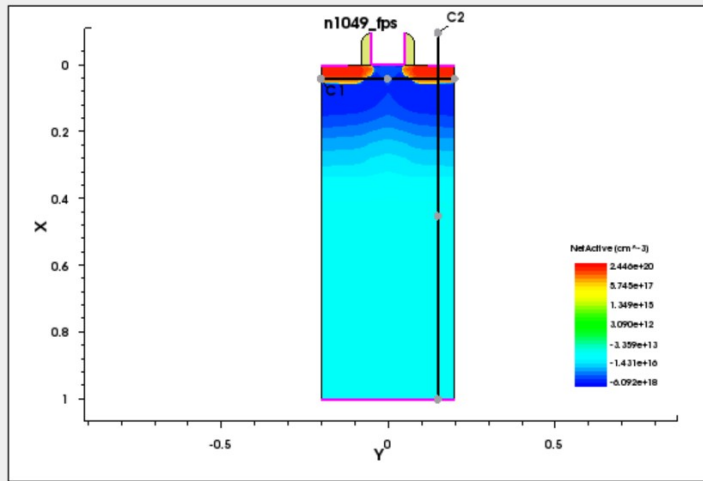
③ Lg 별 Net Active ($X=0.04$, $Y=Lg*1.5Cut$) Profile 그릴것



Lg=0.05
X=0.04 cut
Y=Lg*1.5=0.075 cut

Device 특성

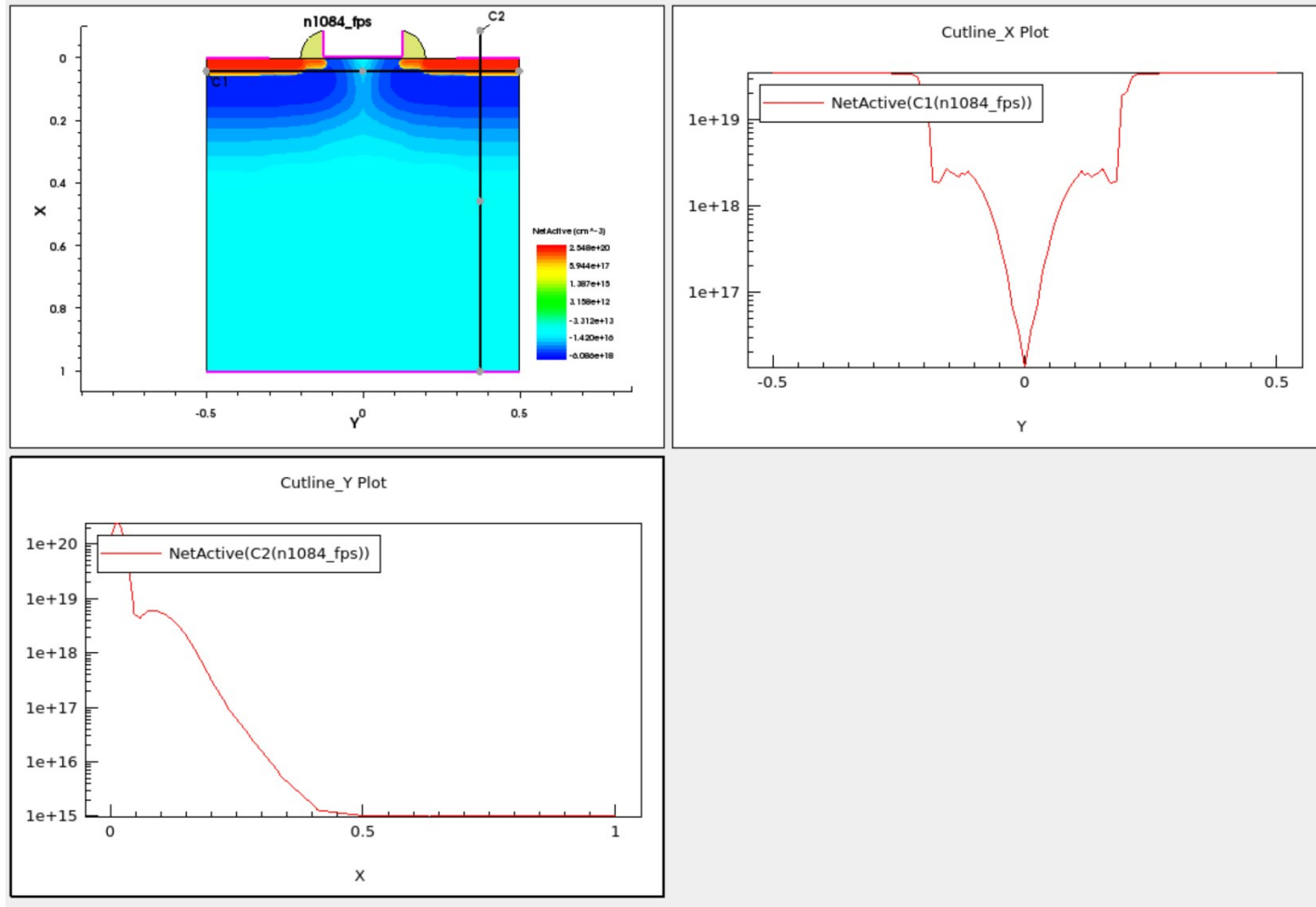
③ Lg 별 Net Active ($X=0.04$, $Y=Lg*1.5\text{Cut}$) Profile 그릴것



Lg=0.1
X=0.04 cut
 $Y=Lg*1.5=0.15$ cut

Device 특성

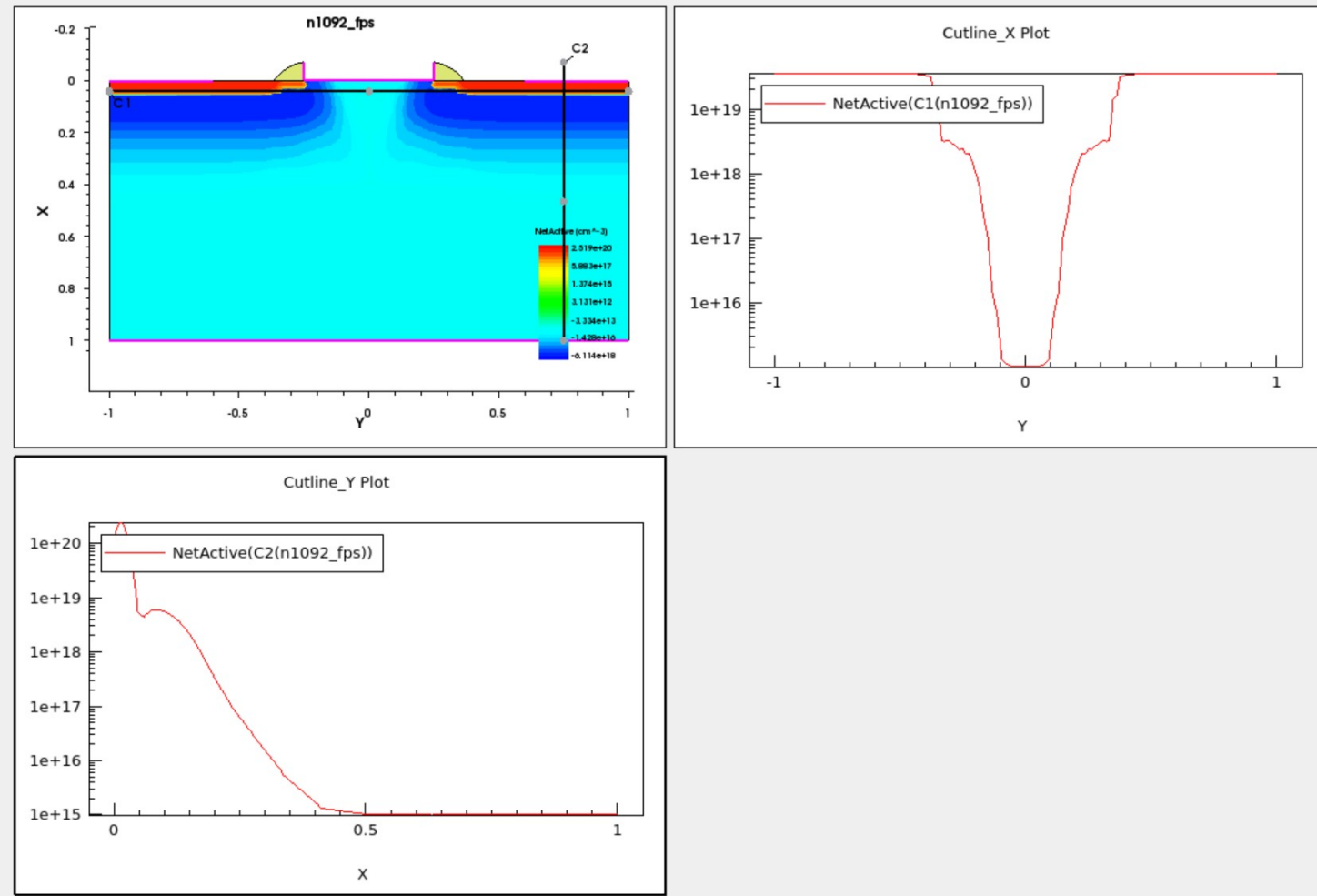
③ Lg 별 Net Active ($X=0.04$, $Y=Lg*1.5\text{Cut}$) Profile 그릴것



Lg=0.25
X=0.04 cut
 $Y=Lg*1.5=0.375$ cut

Device 특성

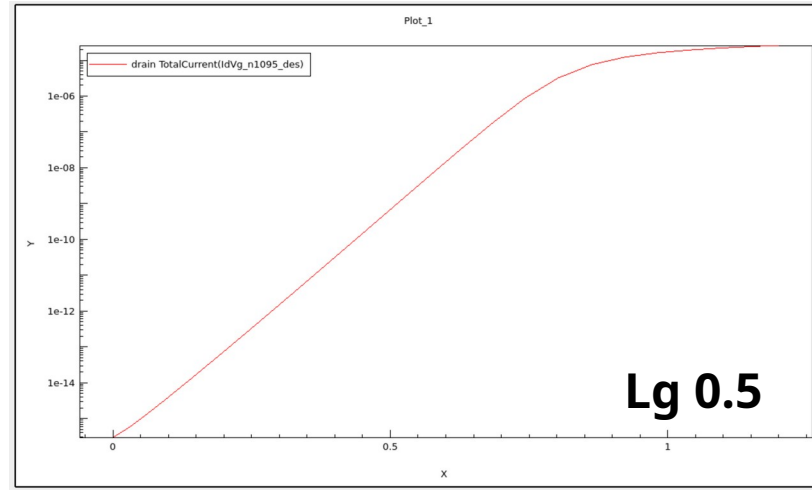
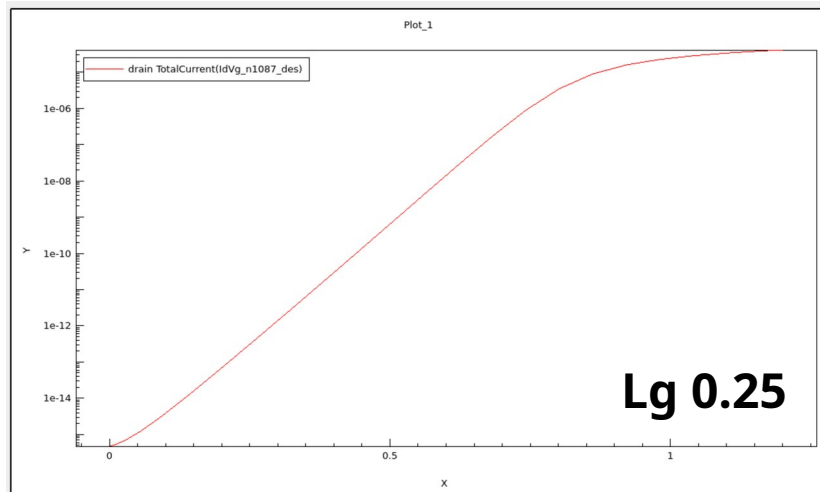
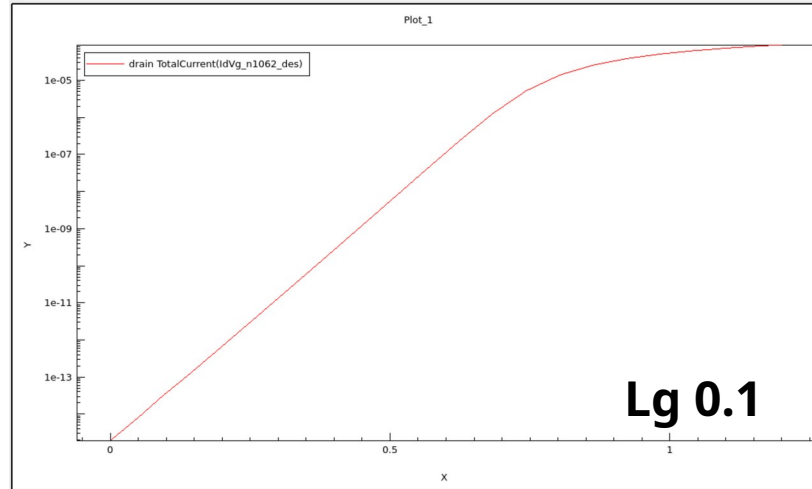
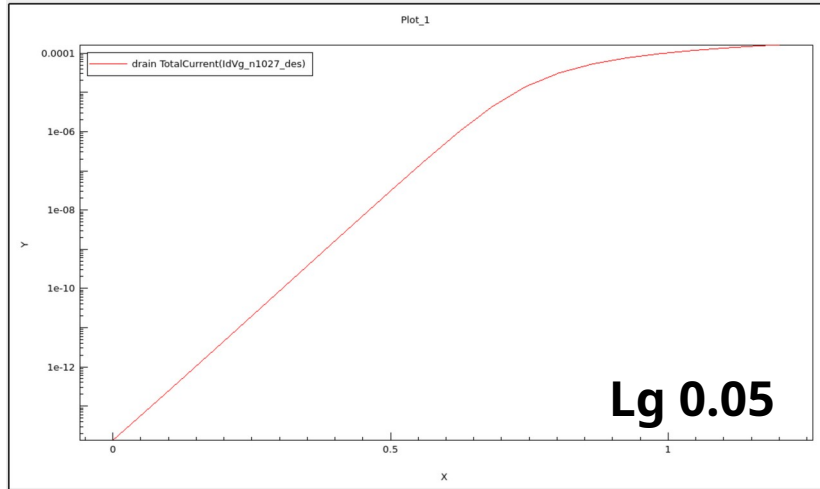
③ Lg 별 Net Active (X=0.04, Y=Lg*1.5Cut) Profile 그릴것



Lg=0.5
X=0.04 cut
Y=Lg*1.5=0.75 cut

Device 특성

④ Lg 별로 Transfer curve 그릴 것 . (@Vd=0.05)



Device 특성

⑤ Lg 별로 On/Off Ratio 표시할 것 (@Vd=0.05V)

Lg	Id	Ioff	Ion/Ioff
0.05	1.64E-04	1.37E-14	1.19E+10
0.1	9.07E-05	1.96E-15	4.63E+10
0.25	4.12E-05	4.73E-16	8.69E+10
0.5	2.59E-05	2.99E-16	8.68E+10

